

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/349140513>

Monitoring Training Loads in Basketball: A Narrative Review and Practical Guide for Coaches and Practitioners

Article in *Strength & Conditioning Journal* · September 2021

DOI: 10.1519/SSC.0000000000000620

CITATIONS

14

READS

2,408

3 authors:



Aitor Piedra

University of Vic

18 PUBLICATIONS 25 CITATIONS

[SEE PROFILE](#)



Javier Peña

University of Vic

77 PUBLICATIONS 693 CITATIONS

[SEE PROFILE](#)



Toni Caparros

National Institute of Physical Education (INEFC)

66 PUBLICATIONS 303 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



eHealth and Physical Exercise Prescription [View project](#)



RECOVERY PROCESS IN SPORT SCIENCES [View project](#)

Monitoring Training Loads in Basketball: A Narrative Review and Practical Guide for Coaches and Practitioners

Aitor Piedra, Msc,¹ Javier Peña, PhD, CSCS*D,^{2,3} and Toni Caparrós, PhD^{1,2}

¹National Institute of Physical Education and Sport of Catalonia (INEFC), University of Barcelona (UB), Barcelona, Spain; ²Sport Performance Analysis Research Group (SPARG), University of Vic-Central University of Catalonia (UVic-UCC), Barcelona, Spain; and ³Sport and Physical Activity Studies Centre (CEEAF), University of Vic-Central University of Catalonia (UVic-UCC), Barcelona, Spain

ABSTRACT

Monitoring training load provides information about the physical demands of the sport in which athletes are competing. Strength and conditioning coaches need to use this information to periodize and make decisions on training to optimize performance and prevent injuries. The following narrative review presents the current state of knowledge on monitoring external and internal loads in basketball. The reviewed articles were classified according to the sample represented, considering 3 groups: elite, subelite, and young basketball players. In addition, we analyzed the recording procedures and methodology used in technological and ecological terms. Articles were classified, considering what was monitored and the context in which it was monitored. These settings included competition,

training, and small-sided games. Performance and sports injury variables were also taken into account in our review.

INTRODUCTION

Training is the process of performing systematic exercises to acquire specific sports skills and improve physical abilities (38). When athletes train properly, exercises induce a functional adaptive response. It is these adaptations that bring about changes in physical performance, injury resistance, or health. Exercise induces a psychophysiological response, and it is this response that provides the stimulus for adaptation (14). A single exercise can generate a stimulus that elicits an acute adaptive response, whereas the systematic repetition of that stimulus and the associated response are necessary to produce chronic adaptations (23). Training stimuli have to be applied over sufficient periods and must be strong

enough to prevent those adaptations from fading before the competition (63).

Basketball is an intermittent sport with short, intense actions, usually less than 3 seconds, and longer periods of moderate activity and recovery (2). Because of this particular nature, and an individual response to the activity, prescribing appropriate training loads seem vital to ensure proper acute and chronic adaptations. Training load has been described as the variable which, appropriately managed, helps to obtain the desired training response (44). The objective of monitoring it is to optimize this process, facilitating decision-making and the reduction of injury risk factors (79). Load is defined as the sum of external load (EL) and internal load (IL) (44). The term “external load” refers to any external stimulus to which

KEY WORDS:

internal load; external load; performance; injury rate; technology

Address correspondence to Javier Peña, javier.pena@uvic.cat.

an athlete is subjected that will give rise to different physiological and psychological responses, after an interaction with other biological and environmental factors (15). The individual response to this interaction, with biological stressors imposed on an athlete during training and competition, is known as “internal load” (90).

Appropriate monitoring of athletes enables coaches to define the possible relationship between load and injury risk. Continuous checking and precise analysis of both sport and nonsport loads can also be a vitally important element in tracking athletes’ performance and their emotional well being (71). Monitoring training can also enhance understanding of training responses, reveal fatigue, detect recovery needs, as well as providing feedback on sports planning, and assisting decision-making in training programs. Keeping a check on load is also important for managing human resources to adapt to competition schedules and ensuring therapeutic load levels that minimize the risk of injury and disease (90).

Measuring EL involves quantifying the training or competition load experienced by an athlete, from variables such as training or competition time (seconds, minutes, hours, or days) (28), training or competition frequency (daily, weekly, or monthly sessions, e.g.) (22), type of training or competition (12), analysis of positional data (with satellite global positioning systems, e.g.) (7), power output/speed/acceleration (47), neuromuscular function (94), number of repetitions of movements (shots, throw-ins and jumps, and changes of direction) (53), and distance (kilometers covered, whether running, cycling, or swimming) (54). Monitoring external factors such as life events, everyday problems, or travel can be equally important (15).

Internal load is measured by evaluating the physiological and psychological response to a given EL (45), and it includes variables such as rating of perceived exertion (RPE) (75), session

RPE (sRPE) (duration of session [min] $\times$ RPE) (24), and psychological questionnaires such as Profile of Mood States (POMS) (62), the Recovery-Stress Questionnaire for Athletes (REST-Q-Sport) (48), Daily Analysis of Life’s Demands for Athletes (DAL-DA) (77), and the Total Quality Recovery (TQR) scale (49). Also considered as factors affecting IL are sleep (quality and duration) (35), biochemical/hormonal/immunological evaluations (57), psychomotor speed (64), heart rate (HR) (37), HR-RPE relationship (56), HR variability (HRV) (69), training impulse (TRIMP) (9), blood lactate concentrations (11), and blood lactate/RPE ratio (89).

Winning in basketball requires the combination of sport-specific training to optimize performance, the improvement of the athlete’s potential in the specific competitive context, and limiting the exposure of players to injury risk scenarios (27). Analysis of these variables is aimed at obtaining information to achieve 2 objectives: maximizing performance and minimizing injury rate. In team sports, better performance is obtained, and the risk of injury is lower, when weekly changes in load are relatively small rather than when large fluctuations in load occur (39). Provided that athletes attain these loads in a gradual and controlled way, high loads and intense physical training seem to offer a protective effect against injuries because of the mediating role of adaptation and the development of physical qualities (18). Excessive and rapid increases in the load to which athletes are subjected, relative to the ability to withstand training loads for which they are prepared, are one of the main factors that can increase the risk of players being injured and reduce their performance (3). Large changes in the application of acute training loads (rapid increases in intensity, duration, or frequency) are related to a higher risk of sports injuries occurring (29). In this context, the relationship between acute and chronic load can be used to model the changes between load and injury

risk (29). Acute load is usually defined as the load from the past week and chronic load as the cumulative load of the past 4 weeks (30). If the chronic load is gradually increased to high levels and the acute load is at a low level, players are well prepared for competition (41). On the other hand, if acute load exceeds chronic load, players are considered to be poorly prepared and at a greater risk of injury (40).

Training loads can be monitored by ecological and technological means. The former do not entail any financial cost, and their reliability has been validated (82). Moreover, the whole coaching staff can help with data collection because the procedures involved in collecting indicators are simple to perform (90). However, technology provides more objective information on the training process, and for this reason, many coaches and players are opting for the use of technological means to monitor their training (35). Training-load monitoring is conducted at all levels of sports practice, although with different degrees of complexity. These processes are performed both in elite teams (players who practice the sport professionally) (74) and in subelite teams (university or minor-league players who supplement the sport with other jobs) (25) and are particularly notable in young players (up to the age of 18), given the strong relationship between high-training volumes in adolescence, injury rates, and abandonment of sports practice (43). In all the cases described above, the load is monitored and managed both in training and in competition (76). There are examples in the literature of monitoring in training situations being applied to the whole training session (92), simulated competition tasks (83), and small-sided games used as means of optimizing play (80). Moreover, with the technological methods now available, the exertion involved in official competition can also be monitored, provided by devices that do not contravene the competition rules and are approved by the relevant players’ associations and the management (68).

Monitoring training loads is an important aspect in the research literature and in the professional sphere. The objective of this study is, therefore, to present a narrative review of the current literature related to monitoring training loads in basketball considering its applicability and regarding the player level, the methodology used, the type of recorded data, and their associations to performance and injury rate.

PLAYER LEVELS

ELITE

In elite basketball, coaches should use teamwork to tailor training sessions as much as possible to competition requirements (4) (Table 1). In this process, appropriate monitoring of EL plays a very important part (72). One of the easiest variables to record, and therefore one of the most widely used, is total session time (93), which provides quantitative information about the stimulus players are receiving (60). In games, recording the number of minutes played (18) and ascertaining the percentage of playing time relative to the total game time enabled coaches to obtain information on each player's contribution (67). However, the most prominent EL recording variables in the literature are the number of accelerations and decelerations ($\geq 2 \text{ m/s}^2$) (97), the distance covered (18), the number of jumps and changes of direction (60,66,92), peak velocity (km/h), and peak acceleration (m/s^2) (97). Another very important measure is known as player load (PL) or body load. This metric is a vectoral quantity expressed as the square root of the sum of the instantaneous rates of change in acceleration in each of the 3 planes, divided by 100 (80). Knowing the different types of movement that occur during play, walking ($<6.0 \text{ km/h}$), jogging ($6.0\text{--}12.0 \text{ km/h}$), running ($12.1\text{--}18 \text{ km/h}$), high-intensity movements ($18.1\text{--}24.0 \text{ km/h}$), and sprints ($>24.0 \text{ km/h}$) also defines the real demands of the sport fairly reliably (67). Elite

players cover lower total distances (between 1,991 and 6,310 m); this could be due to greater movement economy (87). In games, the total number of accelerations ranges between 43 and 145, total high-intensity accelerations between 1 and 15, total decelerations between 24 and 95, and high-intensity decelerations between 4 and 40 (98). In training sessions, the total number of accelerations is between 16.9 and 59.8, high-intensity accelerations between 1.9 and 7.2, total decelerations between 16.4 and 93, and total high-intensity decelerations between 1.6 and 12 (88,92,93). In addition, other variables have also been used in elite basketball, such as mechanical load (which measures a player's change of speed over the course of the game and in which each level includes a weighting factor to account for the severity of the acceleration and deceleration) (4), physiological load (mass·average velocity·distance), physiological intensity (physiological load/minutes played), mechanical intensity (mechanical load/minutes played), walk maximal speed (player individually achieved 0–20% of maximal speed), run maximal speed (player individually achieved 40–60% of maximal speed), sprint maximal speed (player individually achieved 60–80% of maximal speed), maximal speed (player individually achieved $>80\%$ of maximal speed), average offensive speed (average speed of runs in offense [m/h]), average defensive speed (average speed of runs in defense [m/h]) (18), ratio of accelerations to decelerations (98), and total EL (arbitrary units/min) (98).

The most widely used IL variables for elite basketball players are RPE and session sRPE (4,6,26,60,65,66,92,93), to which must be added a series of quantitative psychological questionnaires such as the Recovery-Stress Questionnaire (REST-Q 76) (66), DALDA, and the Wisconsin Upper Respiratory Symptom Survey-21 (WURSS-21) (26). Total weekly sRPE ranges from 2,250 to 5,058 arbitrary units (or AU; a unit based on individual

judgment or preference which is usually used to show the relationship to a predetermined reference measurement); so large a difference could be due to variability in the number of training sessions and their duration (86). From a physiological standpoint, saliva has also been used (60,66,80) and other immunological markers (66). Expressions of HR, including HR (4), Banister TRIMP (4), HRmax (80), and summated HR zones (SHRZ) (65) were very commonly used until the advent of GPS technologies. Maximum HR (HRmax) ranges from 187 to 198 $\text{b}\cdot\text{min}^{-1}$ and average HR (HRavg) is between 150 and 170 $\text{b}\cdot\text{min}^{-1}$ in professional basketball games (50). In addition, the lactate value obtained in games is $5.1 \pm 1.3 \text{ mmol/L}$ (2).

SUBELITE

In subelite-level basketball, the most commonly used EL variable is session duration (13,20,25,36,61,76,81) (Table 2). Numbers of impacts, jumps, and steps are also variables used for this category of player (76), as well as PL (36,76). Four levels of intensity of acceleration and deceleration in subelite basketball have been proposed. Level-1 accelerations (A-1) range from 0.50 to 0.99 m/s^2 , A-2 from 1.00 to 1.99 m/s^2 , A-3 from 2.00 to 2.99 m/s^2 , and A-4 from 3.00 to 50.00 m/s^2 , and level-1 decelerations (D-1) range from -0.50 to -0.99 m/s^2 , D-2 from -1.00 to -1.99 m/s^2 , D-3 -2.00 to -2.99 m/s^2 , and D-4 from -3.00 to -50.00 m/s^2 (8). Other variables that have been recorded are absolute and relative PL and the equivalent estimated absolute and relative distance representing the distance covered at a “jogging” intensity to reach the resulting PL outcome (25). The total distance covered in games ranges from 3,722 to 6,208 m (25,85). Obviously, these variables can be monitored when the appropriate technology is available to do so.

Regarding IL, sRPE is the most commonly recorded variable in subelite-level basketball players (20,55,61,81,86), followed by RPE (20,25,61). In terms of quantitative questionnaires, we find

Table 1
Elite basketball articles reviewed

Reference/Player levels	Internal load	External load	Technology	Type of record	Injury (I)/performance (P)	Conclusions
Vazquez-Guerrero et al. (97)/Elite		Distance (m), player load (AU), peak speed (km/h-1), high intensity actions, accelerations, decelerations, high intensity accelerations of high intensity decelerations, and peak acceleration (m/s ²)	WIMU PRO; realtrack systems SL	Simulated competition		Physical demands can be modulated changing the rules and court size using 5-on-5 scrimmage situations
Svilar et al. (93)/Elite	RPE (CR-10) and sRPE (AU)	Training duration (min), accelerations (n), decelerations (n), jumps (n), and changes of direction (n)	Catapult innovations S5 devices	Basketball training		Acceleration and change of direction for centers, deceleration and high jumps for guards and high and total amount of deceleration and change of direction for forwards are specifically demanded in professional basketball training
Aoki et al. (4)/Elite	Heart rate (b·min-1), Banister's TRIMP, RPE (CR-10), sRPE (AU), physiological load.	Session duration (min), mechanical load (AU), and peak acceleration (m/s ²)	(Bioharness, Zephyr technology, Auckland, New Zealand), photoelectric cell timing system (Multisprint, Hidrofit, MG—Brazil)	Basketball training	(P) Yo-Yo IR1, countermovement (CMJ) and squat jumps (SJ), repeated sprints ability test, and 5'-5' submaximal run	Both eTL and iTL measures should be monitored in association with physical tests, to provide a comprehensive understanding of the training process
Moreira et al. (60)/Elite	Saliva, RPE (CR-10), sRPE (AU)	Session duration (min)		Simulated competition and competition		Session RPE seems to be a viable tool in monitoring internal loads, and the results are useful in providing a better understanding of internal loads imposed by basketball training and competitions

Table 1
(continued)

Nunes et al. (66)/Elite	RPE (CR-10), sRPE (AU), the recovery-stress questionnaire for athletes (RESTQ76), salivary hormone and immune markers	Session duration (min)	Two electronic time sensors (Photo-cell system; CEFISE, Nova Odessa, SP, Brazil), a jump mat (Ergojump jump Pro 2.0; CEFISE)	Basketball training	(P) Bench press, squat, Agility t Test, the Yo-Yo intermittent recovery test, level 2, and squat jump test	Session RPE and REST-Q can provide useful low-cost information for monitoring both ITL and stress state in an elite athlete group.
Sansone et al. (80)/Elite	HR max ($b \cdot \min - 1$), saliva, Edwards' TL (AU)	Player load (AU) is calculated as the instantaneous rate of change in accelerations in the anteroposterior, mediolateral, and craniocaudal axes	Catapult OptimEye S5, Polar H7	Small sided games	(P) Yo-Yo intermittent recovery test (level 1)	Cortisol increased after the SSGs, whereas testosterone responses differed depending on SSGs conditions.
Caparrós et al. (17)/Elite		Minutes played, physiological load, physiological intensity, mechanical load, mechanical intensity, distance covered (mL), walking maximal speed, run maximal speed, sprinting maximal speed, maximal speed, average offensive speed (mL/h), average defensive speed (mL/h), level 1 acceleration (0.5 m/s ²), level 2 acceleration (1 m/s ²), level 3 acceleration (2 m/s ²), level 4 acceleration (3 m/s ²), level 1 deceleration (−0.5 m/s ²), level 2 deceleration (−1 m/s ²), level 3 deceleration (−2 m/s ²), and level 4 deceleration (−3 m/s ²)	Player tracking.	Competition	(I) Based on the methodology of the UEFA consensus statement for epidemiological studies (reference by #) (P) Player efficiency rating (PER) and usage percentage (Usg%)	Increasing external workload may likely reduce the risk of injury in professional basketball

(continued)

Table 1
(continued)

Svilar et al. (92)/Elite	RPE (CR-10) and sRPE (AU)	Session duration (min), player load (PL), total and high acceleration/ deceleration, total jumps and jumps performed within the high band: tCoD (total inertial movements registered in a rightward lateral vector), and hCoD (total inertial movements registered in a rightward lateral vector within the high band)	S5 devices (Catapult Innovations, Melbourne, Australia)	Basketball training		Triaxial accelerometry and sRPE can be used complementary, when possible, to design and control the training process as effectively as possible. An individualized approach to external load monitoring in basketball is a complementary tool that could help coaches and teams minimize injuries while achieving the best performance
Nunes et al. (65)/Elite	SHRZ (AU), RPE (CR-10), and sRPE (AU)		Polar team system	Competition	(P) The game performance indicators used in this study were: game time, number of shots thrown (3 pts, 2 pts, and free throws), number of converted shots (3 pts, 2 pts, and free throws), defense and attack rebounds, assists and recovered balls	Internal load estimated by the Edwards method was correlated with the technical indicators of game performance (number of points scored, rebounds, assistance, and recovery of ball possession). Number of actions performed in the match by the athletes influences the internal load. Monitoring the internal load of official matches can help the coaching staff to plan
Vázquez-Guerrero et al. (98)/Elite		Accelerations (<3 m/s ²), accelerations (>3 m/s ²), decelerations (<3 m/s ²), decelerations (>3 m/s ²), acceleration: deceleration ratio (<3 m/s ²), acceleration: deceleration ratio (>3 m/s ²), and external total load (AU/min)	Triaxial accelerometers (model ADXL326; Analog devices, Inc., Norwood, MA)	Competition		More maximal decelerations than accelerations were performed in all playing positions during elite basketball competition games. The acceleration: deceleration ratio (>3 m/s²) was significantly lower in players on the perimeter than in PF and C.

Table 1
(continued)

Oliveira-Da-Silva et al. (67)/Elite		Low-intensity actions: walk ($<6.0 \text{ km} \cdot \text{h}^{-1}$), jog ($6.0\text{--}12.0 \text{ km} \cdot \text{h}^{-1}$), and run ($12.1\text{--}18 \text{ km} \cdot \text{h}^{-1}$). Low-intensity shuffle (n), medium-intensity shuffle (n). High-intensity actions: high-intensity displacements ($18.1\text{--}24.0 \text{ km} \cdot \text{h}^{-1}$) and sprints ($>24.1 \text{ km} \cdot \text{h}^{-1}$), high-intensity shuffle (n), jump (n). Total time on court (s), real time (s) and percentage of TR with respect to TT (TR/TT)	Recordings used were provided by FIBA and stored on a laptop. Films were analyzed frame by frame with a precision of 0.04 s	Competition		The use of specific basketball actions, adapted in intensity and volume to real demands, as a means of training guarantees the specificity of the stimulus
Freitas et al. (26)/Elite	RPE (CR-10) and sRPE (AU), DALDA, and WURSS-21			Basketball training		A decrease in ITL during the competitive period was followed by a decrease in stress tolerance and an increase in URTI severity.
Arruda et al. (6)/Elite	RPE (CR-10) and sRPE (AU)			Basketball training		The higher correlation coefficient between AD and CIT-A suggests that the predicted early season (PD) difficulty should be updated during the competitive season
Caparrós et al. (17)/Elite		Exposure time (h)		Basketball training and competition	(I) Based on the methodology of the UEFA consensus statement for epidemiological studies (reference #) (P) Performance statistics were collected from the official game statistics for both the team and player. At the end of each game, the official ranking (RKG)	Although increasing practice and competition time is related to greater team performance, it also increases the number of injuries. However, higher injury rates were not associated with worse overall team performance

(continued)

Table 1
(continued)

Schelling and Torres (88)/Elite	Acceleration data (AL · min – 1)	Triaxial accelerometer (X8-mini; 16-bit; Gulf Coast Data Concepts)	Basketball training	Full-court 3v3 and 5v5 showed the highest external workload, measured by triaxial accelerometer. The smaller the player, the higher the acceleration load,
---------------------------------	-------------------------------------	---	------------------------	--

TQR (81) and Wellness questionnaires (61) used with this category of player. The different expressions of the HR are also used: HR (10), TRIMP (36,86), HRavg, HRmax, %Maximum HR (% HRmax) (76), and SHRZ (13,25,76,86). It has also been observed that the use of SHRZ 2.5 at this level of competition provides a novel view of IL and may be more sensitive for detecting nonadaptive and adaptive responses to training (84). In games, HRmax ranges from 192 to 195 $\text{b} \cdot \text{min}^{-1}$ and HRavg is between 168 and 169 $\text{b} \cdot \text{min}^{-1}$ (50). The use of noninvasive strategies to monitor internal stress is a valuable tool for identifying fatigue and stress in subelite-level players (36).

YOUNG PLAYERS

For young basketball players (under 18 years old), session duration (21,51,52,58) predominates over the other indicators used to monitor training (Table 3). The number of minutes played during the game is also frequently recorded (95). Total distance covered (34,42,91), relative distance (m/min) (5), and distance covered at high intensity (speed greater than 16 km/h) (34), PL and total number of accelerations and decelerations (5,34), as well as frequency of accelerations and decelerations (91) could also be indicators used in this population when there are means available to do so. In games, the distance covered is approximately 7,558 m in young players (1). The numbers of impacts of low intensity (3–5 g), medium intensity (5–8 g), and high intensity (>8 g) and of jumps and steps (33,34,73) have also been used by various researchers to ascertain the demands of competition in this category. Other variables that have been recorded in young players include measurement of mean sprint speeds (83), high intensity running (HIR: percentage of total distance covered at over 16 km/h) (68), peak velocity (km/h) and peak acceleration (m/s^2) (5,68), velocity (m/s) (42), work-rest ratio (42), and metabolic power (W/kg) (34).

As regards IL, RPE (21,33,51,52,58,78,91,95,96) and sRPE (21,33,51,96) are again the metrics

used most frequently when it comes to young basketball players. The quantitative questionnaires most commonly used in this type of population are Delayed-Onset Muscle Soreness (DOMS) (5), TQR (21,95), POMS (58), Wellness, and the Postgame Subjective Performance Evaluation (95). Lactate (91) and saliva (58) have also been analyzed in young players, and this information confirms that training programs can be individualized to facilitate better adaptation to training loads. The following expressions of the HR have been used for young players: SHRZ (42,51,52,73), HR (73,91,96), TRIMPs and Lucia TRIMP (83), HRavg, HRmax, %HRmax (73) and HRpeak, HRmean, %HRmax, Z4, and Z5 (78). Maximal intensity (session duration at 90–100% HRmax) has also been analyzed to find out how much time is spent at high levels during competition so as to be able to adapt it to training (52). In games, mean HRmax is 199 $\text{b} \cdot \text{min}^{-1}$ and HRavg is between 167 and 172 $\text{b} \cdot \text{min}^{-1}$ (50).

DATA RECORDING

TECHNOLOGICAL MONITORING

In recent years, monitoring training loads has acquired greater importance for professional athletes (16). The technology used is constantly evolving, and the most common EL variables currently recorded are power output, speed, acceleration, time-motion analysis (TMA), global positioning system (GPS) parameters, and parameters recorded using accelerometry. As for IL measures, the most frequent include HR, oxygen uptake, and lactate concentration (70).

We found that the most widely used device for monitoring EL and IL is the Polar Team2 Pro (42,79,83,86,91). For tracking EL, the WIMU PRO (5,13,34,73,76,97) and the Catapult OptiEye S5 (25,36,80,84,92,93) are most often mentioned (although the former can also be used to monitor IL). However, we found other articles whose data were obtained through optical player tracking systems (18), the ADXL326 accelerometer (98), the

Table 2
Subelite basketball articles reviewed

Reference/Player levels	Internal load	External load	Technology	Type of record	Injury (I)/Performance (P)	Conclusions
Scanlan et al. (86)/Subelite	sRPE (AU), TRIMPS, and SRHZ (AU)	External training load was calculated using an established algorithm developed by Catapult Innovations (Scoresby, Australia)	Polar Team2 Pro HR monitors	Basketball training		Do not assume a linear dose-response between accelerometer and internal training load models during training and are recommended to combine internal and external approaches when monitoring training load in players
Heishman et al. (36)/Subelite	Internal stress monitoring, TRIMP	PlayerLoad (AU), practice duration (min)	Omegawave technology, Catapult Optim Eye S5	Basketball training	(P) Countermovement vertical jump (CMJ)	Using the noninvasive strategies to monitor internal stress and external training load may be valuable tools in identifying fatigue and player stress. These technologies quantify training loads imposed on an athlete, which allows for potential modulation of the training process to optimize athlete performance. In addition, internal stress and external load monitoring may provide different insights into athlete training load and adaptation
Sansoneet al. (81)/Subelite	sRPE (AU), total quality recovery (TQR) scale	Practice duration (min)	Photoelectric cells	Basketball training and competition	(P) Yo-Yo intermittent recovery level 1 (YO-YO IR1), counter movement jump (CMJ), lane agility and 505 agility tests, and 20-m sprint	Utilization of the sRPE method for measuring iTL, calculation of the ACWR, and recovery assessment through the TQR scale can be a valid combination for monitoring daily, weekly, and mesocycles training and recovery
Conte et al. (20)/Subelite	RPE (CR-10) and sRPE (AU)	Total duration (min)		Basketball training and competition		Basketball coaches should concurrently consider the number of weekly games and player status (starting versus bench player) when creating individualized periodization plans

(continued)

Table 2
(continued)

Berkelmans et al. (13)/Subelite	SHRZ (AU), HRmax ($b \cdot \text{min} - 1$)	Session durations (min)	Heart rate monitors (Polar T31, Polar Electro; Kempele, Finland)	Basketball training	(P) Yo-Yo (IRT Yo-Yo)	Age-predicted, test-derived, and session-based methods to determine HRmax are not interchangeable when calculating SHRZ. Use individualized HRmax directly measured during field-based tests supplemented with higher HR responses evident during training sessions and games when calculating the SHRZ TL
Fox et al. (25)/Subelite	RPE (CR-10), sRPE (AU) absolute and relative SHRZ (AU)	Absolute and relative PlayerLoad (AU), practice duration (min), absolute and relative estimated equivalent distance (m)	Microsensors (OptimEye S5, Catapult Innovations, Melbourne, Australia) and chest-worn HR monitors (Polar T31, Polar Electro, Kempele, Finland)	Basketball training and competition		Training demands exceeded competition demands
Moreno-Pérez et al. (61)/Subelite	RPE (CR-10), sRPE (AU), and Wellnes	Session duration (min)	Ergo tester (Ergo jump Bosco system, Italia), LegMotion	Competition	(P) Ankle dorsiflexion and countermovement jump (CMJ)	Ankle dorsiflexion ROM is still reduced 48 h after a competitive basketball match in semiprofessional basketball players.
Scanlan et al. (84)/Subelite	(a) SHRZ10—traditional approach proposed by Edwards using 10% HRmax relative zone ranges; (b) SHRZ5—modified approach using 5%HR max relative zone ranges; and (c) SHRZ2.5—modified approach using 2.5% HRmax relative zone ranges		Polar T31, Polar Electro; Kempele, Finland, OptimEye S5; Catapult Innovations; Melbourne, Australia	Basketball training		Use of SHRZ2.5 provides novel insight regarding internal loading in basketball players and may carry greater sensitivity for detection of maladaptive and adaptive responses to training

Table 2
(continued)

Sánchez Ballesta et al. (78)/Subelite		At the level of accelerations and decelerations per minute: A-1 encompass between 0.50 and 0.99 m/s ² , A-2 between 1.00 and 1.99 m/s ² , A-3 between 2.00 and 2.99 m/s ² , A-4 between 3.00 and 50.00 m/s ² , D-1 between -0.50 and -0.99 m/s ² , D-2 between -1.00 and -1.99 m/s ² , D-3 between -2.00 and -2.99 m/s ² , and D-4 between -3.00 and -50.00 m/s ² . Total-A (87)	Polar team Pro	Basketball training		Levels of approach III and IV show a relationship with external load and could validate the distribution of tasks and external load proposed by the levels of approach in Schelling and Torres-Ronda (87). Opposition may have a relationship in the number of high-intensity accelerations and decelerations per minute.
Marcelino et al. (55)/Subelite	RPE pre and post (CR-10), sRPE (AU)			Basketball training	(P) Yo-Yo intermittent Endurance, Test nivel 2 (Yo-Yo IE2)	The results of this study suggest the existence of a strong correlation between the ability to perform high intensity effort (HIE) and ITL, thus indicating the importance of adopting an integrated monitoring of these variables, aiming to conduct regular adjustments on training load

(continued)

Table 2
(continued)

Batalla Gavalda et al. (10)/ Subelite	FC (b·min ⁻¹)		Suunto TeamPack, 2 video-cameras (JVC-GZ620ES 60 GB; HDD Hong Kong, Xina, Kinovea)	Competition	(P) Incremental field test	The conclusions of this study suggest that the difference in points on the scoreboard and play actions lasting 17–24 s have a direct impact on HR, which changes as a function of the points difference and influences the physiological demands on the players
Reina et al. (73)/ Subelite	Average heart rate (HR avg), maximum heart rate (b·min ⁻¹), % maximum heart rate (%HR max), and work zones. The work zones were Z1 (50–60%), Z2 (60–70%), Z3 (70–80%), Z4 (80–90%), Z5 (90–95%), and Z6 (>95%)	Impacts (higher than 5 Gs), jumps (more than 400 milliseconds of flight time before landing again), player load (AU), time (min)	Garmin heart rate Band and a WimuTM inertial device	Basketball training and competition		The players must work in higher areas of the heart rate during training, mainly in Z4 and Z5, increasing their HR max y HR avg. The training does not equal the load supported and the distance performed in competition, so it is necessary to pay more attention during training

X8-Mini accelerometer (88), and the VERT accelerometer (33). On the other hand, for monitoring IL we found practical evidence obtained using HR sensors such as Garmin™ (73,76), Suunto Memory Belts (10,96), Polar H7 (51,52,80), Polar T31 (25,84), and other models from Polar Team Solutions (65,78). Lactate analyzers (83,91) and Omegawave technology (36) have also been used. Omegawave technology evaluates the athlete's internal stress, readiness to perform, and neurological function through HR variability assessments (36) (Table 4).

There are currently numerous applications that help us manage data relating to training sessions, sports performance, physical and medical tests, and administrative matters in a single solution. Furthermore, these applications enable us to manage wellness variables more easily. Athlete Monitoring (32), Readiness (102), Elite HRV (99), Quanter (100), and MyCoach RPE (101) are examples of some of those we can find on the market.

Another type of technology is also used to analyze players' performance or obtain useful data for the research study we have conducted. We found photoelectric cells (66,81), electronic timing lights (83), the Tanita BC-601 body composition analyzer (34), and contact mats (21,61,66,95). With a counter-movement jump (CMJ) using contact mats, we can detect our players' neuromuscular fatigue and find out how ready they are to begin the session and also discover the differences at the end of the session (5,19,21,31,46,59). In addition, video cameras (10,42,67,78) and video analysis software such as Kinovea (10) and Match Vision (78) have been used.

ECOLOGICAL MONITORING

Good monitoring of training loads does not necessarily have to be conducted using technology. Several ecological methods are available, defined as those that entail little or no cost, proven also to be useful and reliable (82).

Table 3
Young basketball articles reviewed

Reference/Player levels	Internal load	External load	Technology	Type of record	Injury (I)/Performance (P)	Conclusions
Scanlan et al. (83)/Young basketball	sRPE (AU), SHRZ (AU), TRIMP, and Lucia TRIMP models	Measurement of mean sprint and circuit speeds	Polar Team2 Pro HR monitors, electronic timing lights, held lactate analyzer	Simulated competition		Limited convergence exists between common training load approaches across basketball training doses lasting beyond 20 min. Internal and external training load approaches appears dose-dependent during basketball activity
Arede et al. (5)/Young basketball	RPE (CR-10), muscle soreness (DOMS)	(a) Relative distance (RD, m/min); (b) high-intensity running (HIR, % of total distance covered at over 16 km/h); (c) player load, accumulated accelerometer load in 3 axes (PL, AU/min); total accelerations (Acc, count/min) and decelerations (Dec, count/min) and peak speed (PSpeed, km/h) and peak acceleration (PAcc, m/s ²)	WIMU PRO system	Basketball training and competition	(P) Performed physical (jumping, agility, aerobic fitness, and lower limb asymmetry index [ASI]) tests and game-related statistics were also collected	Game-related statistics obtained in FM seem to be a determinant aspect that increases the probability to be the starter during the EC
Vaquera et al. (96)/Young basketball	Individual HR and RPE (CR-10)		Suunto Memory Belts	Small-sided games		The number of players active in small-sided basketball games influences both the physiological demands and the in-task RPE. There are significant discrepancies between athlete and coach perceptions of in-task RPE and therefore of internal training load imposed

(continued)

**Table 3
(continued)**

Cruz et al. (21)/ Young basketball	sRPE (AU), RPE (CR-10), TQR scale	Session duration (min)	Contact platform (Sys Jump, Systware—Brazil)	Basketball training	(P) CMJ	Maintenance of the balance between recovery and stress allows meaningful improvements in physical performance. Internal training loads experienced across competitive periods can help to appropriately define and adjust their training programs
Lupo et al. (51)/ Young basketball	sRPE (AU), RPE (CR-10), EDWARDS TL (AU)	Session duration	Polar H7	Basketball training		Coaches of youth basketball players can successfully use session-RPE to monitor the ITL, regardless of session durations and workout portions
Pino-Ortega et al. (68)/Young basketball		(a) Relative distance (RD, m/min); (b) high-intensity running (HIR, % of total distance covered at over 16 km/h); (c) Player load, accumulated accelerometer load in the 3 axes of movement (PL, AU/min); total accelerations (Acc, count/min) and decelerations (Dec, count/min) and PeakSpeed (PSpeed, km/h) and peak acceleration (PAcc, m/s ²)	WIMUPROTM, Real Track systems, Almería, Spain	Competition		Study the opponent to design the physical load demands in training sessions according to quality level; (b) guards and forwards should cover higher distances and reach greater PSpeed and PAcc than centers during training to simulate the match physical requirements; (c) to avoid the effect of fatigue, interperiod recovery strategies should be arranged by medical staff and total playing time should be monitored and distributed among the players(d) Prescribe correct physical, tactical, and technical demands during the precompetition period to achieve the best performance from the first match of the tournament

Table 3
(continued)

Miloski et al. (58)/ Young basketball	Testosterone concentration Assay (saliva), RPE (CR-10), sRPE (AU), profile of mood states	Session duration (min)		Basketball training	(P) Yo-Yo IRL1, t Test	LTC athletes may be more susceptible to changes in mood states during intensified training periods. Data indicate that a periodized training program successfully improved the physical performance (endurance and agility) of young basketball players
Stojanović et al. (91)/Young basketball	Heart rate (b·min ⁻¹) responses, RPE (CR-10), and blood lactate concentration (BLa)	Total distance covered (m) and the frequency of accelerations and decelerations acceleration and deceleration intensities are low (0.5–0.99 m/s ²), medium (1–1.99 m/s ²), and high (>2 m/s ²)	Polar team Pro, lactate Scout	Small-sided games		Basketball SSG consisting of 1–3 players elicits an intermittent activity profile, promoting the extensive utilization of anaerobic and aerobic metabolism. SSG with only 1 player per team elicits higher BLa, RPE, distances covered at low speeds as well as acceleration and deceleration volumes than SSG with 3 players per team
Lupo et al. (52)/ Young basketball	RPE (CR-10), sRPE (AU), Edwards' data (AU), maximal intensity (session duration at 90–100% HRmax)	Total session duration (min), type of training (strength, conditioning, and technique)	Polar H7	Basketball training		Elite youth female basketball players are better able to quantify the internal training load of their sessions than their coaches, strengthening the validity of s-RPE as a tool to monitor training in team sports
Gómez-Carmona et al. (34)/ Young basketball		Total distance (m), distance at high intensity (m at more than 16 km/h), accelerations and decelerations (n),	WIMU PRO TM, body composition monitor model BC-601	Competition	(P) The score difference in each of the disputed quarters, classifying into the disputed match (difference less than 10 points) and	The difference in points at the end of each game period/ is a determining factor in the external load. Therefore, it is necessary to reproduce these (continued)

Table 3
(continued)

		impacts at different ranges (3–5, 5–8, > 8 G), steps (n) and jumps (n), player load (AU), and metabolic power (W/Kg)	(TANITA, Tokyo, Japan)		unbalanced match (difference greater than 10 points)	situations during training to subject players to physical–technical–tactical stimuli of equal or greater than volumes and intensities in competition
Reina et al. (73)/Young basketball	Heart rate ($b \cdot \min - 1$), average heart rate (HRAvg); maximum heart rate (HRmax), % maximum heart rate (% HRmax) and work zones, are divided according to the percentage of maximum heart rate that each task implies individually, being Z1 (50–60%), Z2 (60–70%), Z3 (70–80%), Z4 (80–90%), Z5 (90–95%), and Z6 (>95%)	Number of impacts, number of steps, and number of jumps	GARMINTM brand heart rate Band and a WIMUTM inertial device	Competition		Higher demand was found in IC and EC in 3-on-3 basketball
Vallés-Ortega et al. (95)/Young basketball	RPE (CT-10), TQR, perceived well-being questionnaire, Subjective assessment of postmatch performance	Game minutes, CMJ	Chronojump-Boscosystem jump platform	Competition		The resulting data shows the possibility of knowing and reliably monitoring fatigue and recovery in women's basketball during a period of highly competitive density.

Table 3
(continued)

Hůlka and Cuberek (42)/ Young basketball	Heart rate every 5 s (b·min ⁻¹), HR peak zones (AU)	Distance covered (m), speed (m/s), work: rest ratio)	TEAM Polar2 Pro, 2 Canon HF10 video cameras. Video records were analyzed by a standardized procedure using a software package	Competition		Combination of heart rate and time motion analysis is recommended
Sánchez-Sánchez et al. (78)/ Young basketball	RPE (CR-10), HR peak, HR mean, % HR max, Z4, and Z5		HR monitors (Polar team system 2, Polar Electro OY, Finland), video-camera (Sony Handycam HDR-PJ240E, SONY China), observational ad hoc software (Match Vision)	Small-sided games	(P) YoYo IR1. The technical demands analyzed after performing SSG were the defensive rebounds, offensive rebounds, total succeed passes, total number of 3 point shots, total number of 2 point shots, number of performed lay-outs, total number of possessions, success in actions with the possibility to score points, and total number of official stops	SSG with verbal encouragement should be selected to focus in physiological load, and SSG without dribbling would be particularly useful to enhance collective behavior, without limitations in physiological load
Ghali et al. (33)/ Young basketball	RPE (CR-10) and sRPE (AU)	Jump WL (i.e. jump count) and session duration (min)	VERT Classic accelerometer (version 2.0, Mayfonk Inc., Fort 161 Lauderdale, FL)	Basketball training and competition		A significant difference in WL was not detected between youth basketball players with patellar tendinopathy and players with no patellar tendinopathy. Players at early stages of patellar tendinopathy and applying relevant interventions are warranted

Table 4

Distribution of technology according to its monetary cost and the type of recorded variables

Technology	Cost	External load	Internal load
Polar team 2 Pro	***	X	X
Catapult OptimEye S5	****	X	
WIMU PRO	*****	X	
Player tracking	****	X	
Accelerometer ADXL326	***	X	
Accelerometer X8-mini	**	X	
Accelerometer VERT	*	X	
GarminTM	*		X
Suunto Memory Belts	*		X
Polar H7	*		X
Polar T31	*		X
Lactate analyzer	***		X
Omegawave technology	****		X

Technology: technology that has been used to monitor training loads; cost: monetary cost orientation of each technology *low **moderate ***medium ****high *****very high; external load: external load variables; internal load: internal load variables.

RPE is a cost-free ecological tool that gives us information about how the player perceived the stimulus of training or competition (51,52). It is normally used between 10 and 30 minutes after finishing training (21,25,66). In some studies, RPE is administered before and after the session to assess fatigue (55). In addition, sRPE seems to be a viable tool for monitoring the IL imposed by basketball training and competition (60). When RPE is multiplied by session duration, the result is influenced more by the volume of the session than by the intensity of the training (52). Being so easy to use, it is the method most commonly used to monitor load (6,26,33,61,65,91).

A chronometer can be used to measure session duration, providing us with an important piece of EL information (20,36). It is important to know the difference between real time and total session time because this will change the load to which players are subjected (67). This parameter can very

easily be collected in a notebook or a computer document (13,21,52,61,92). By recording time on the court, coaches may modify the training program accordingly (95).

The quantitative questionnaires that have most often been used in basketball are the TQR (21,81,95), the REST-Q-Sport (66), the POMS questionnaire (58), the Wellness questionnaire (61,95), the Postgame Subjective Performance Evaluation (95), DALDA, and the Wisconsin Upper Respiratory Symptom Survey-21 (WURSS-21) (Table 5).

Other high-level ecological tools for assessing the state of basketball players are the DOMS scale (5) and the type of drill used in training. Assessing and monitoring the tasks used in daily practices can help to understand synergies with other aspects of athletes' preparation to obtain performance (52).

TYPE OF REGISTER

Monitoring load during competition provides us with information about

what happens on the court and tells us how to adapt the training process to the demands of competition (68). During a game, more maximal decelerations than maximal accelerations occur in all playing positions. The acceleration:deceleration ratio ($>3 \text{ m/s}^2$) is significantly lower in perimeter players (point guards and small forwards) than in inside players (power forwards and centers) (98). The points difference at the end of each playing period is also a decisive element in the EL that players are subjected to during games (34), and furthermore, it has been observed that this difference in score has an important impact on physiological demands (10). In competition, a combination of analysis of the HR and of the movements made by players is recommended (42). Monitoring EL can help the coaching staff to plan and program future training loads (65), and these situations need to be reproduced in training to subject players to physical, technical, and tactical stimuli of a volume and intensity equal to or greater than those that occur in competition (34). Maximal deceleration movements should be emphasized in perimeter players and total EL reduced in inside players to best prepare players for game demands (98), and moreover, it has been seen that using specific basketball actions, adapted in intensity and volume to real demands, should be the main means of training because this ensures the specificity of the stimulus (67). The coaching staff must study the opponent to design the physical load, which requires training sessions according to its quality level (a higher-level opponent: more volume of demands and a lower-level opponent: more high-intensity actions) (68).

In some research studies, competitions have been simulated to understand what happens in play. It has been observed that physical demands can be modulated by changing the rules and court size, and this factor should be taken into account when designing training drills and when fine-tuning periodization (97).

In training sessions, it can be seen that acceleration and change of direction

Table 5
Abbreviations and definitions

Abbreviations	Definition
EL	External load
IL	Internal load
RPE	Rating of perceived exertion
sRPE	Session RPE
POMS	Profile of mood states
REST-Q-Sport	Recovery-stress questionnaire for athletes
DALDA	Daily analysis of Life's demands for athletes
TQR	Total quality recovery
HR	Heart rate
HRV	Heart rate variability
TRIMP	Training impulse
WURSS-21	Wisconsin Upper Respiratory Symptom Survey-2
AU	Arbitrary units
SHRZ	Summated heart rate zones
HRmax	Maximum heart rate
HRavg	Average heart rate
PL	Player load
%HRmax	%Maximum heart rate
HIR	High intensity running
DOMS	Delayed-Onset muscle Soreness
TMA	Time-motion analysis
GPS	Global positioning system
CMJ	Counter-movement jump
Z4	80–90% heart rate
Z5	90–100% heart rate
YO-YO IR1	Yo-Yo intermittent recovery test level 1
SJ	Squat jump
YO-YO IR2	Yo-Yo intermittent recovery test level 2
ASI	Lower limb asymmetry index
PER	Player efficiency rating
Usg%	Usage percentage

are the most decisive variables for centers, deceleration and high jumps for point guards, and total deceleration and changes of direction for small

forwards (93). It should also be borne in mind that the number of high-intensity accelerations and decelerations that occur per minute may be

related to the opposition. However, we must take account of the fact that playing without opposition, with a lower-quality EL, can be equally intense in terms of high-intensity accelerations and decelerations as playing with opposition (5×5) half-court or full-court, or even more so (79). **Playing 3×3 or 5×5 full court are the exercises that show the highest EL** (88).

Players must work in high heart-rate zones during training, mainly in Z4 (80–90%) and Z5 (90–100%) (76), and this can be achieved using interval exercises without full recovery, to work at the appropriate threshold (76). In basketball, the demands of training usually exceed those of competition (25). **Shooting guards and small forwards experience higher loads than the rest of the team, which should be taken into account when fine-tuning the training program for these positions** (76). Moreover, the smaller the player, the higher the acceleration load they are subjected to (88).

The use of small-sided games to prepare basketball players for competition seems to show benefits in performance (80). Games consisting of 1–3 players give rise to an intermittent activity profile, promoting extensive utilization of anaerobic and aerobic metabolic pathways (91). The 2×2 format seems to be the one with the highest load. Moreover, playing small-sided games with stimuli generates a higher physiological load in players (78).

PERFORMANCE AND INJURY RATE

To evaluate performance in basketball, the measure most commonly used in the literature is the CMJ as a jump test (5,20,21,36,61,81) and the Yo-Yo Intermittent Recovery test level 1 (YO-YO IR1) as an aerobic test (4,13,58,78,80,81). The squat jump (SJ) (4,66) and the Yo-Yo Intermittent Recovery test level 2 (YO-YO IR2) (55,66) have been used to a lesser extent. Agility tests have also been used, such as the Lane Agility Test (81), the 505 Agility test (5,81), and the *t* test (58), and speed tests such as the 20-m sprint (36) or the repeated sprint test (4). Other aerobic tests that have been

Monitoring Training Loads in Basketball

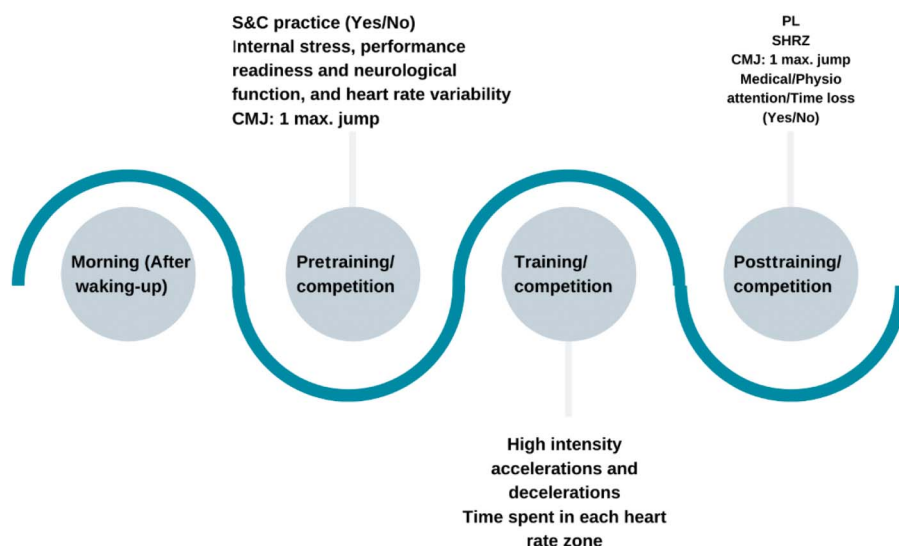


Figure 1. Suggested timeline for technological load monitoring.

used are the 5'-5' submaximal run (4,5) and incremental tests (55). Finally, we have found that other performance measures have been used, such as bench press and squat values (66), the lower limb asymmetry index (ASI) (5), or ankle dorsiflexion (61). On the other hand, game statistics and the values they provide for each player have also been used to ascertain the players' performance (5,66,78). In addition, advanced statistical values such as the player efficiency rating (PER) and

usage percentage (Usg%) have also been used (18), as well as the points difference in the score, which has also been considered as a performance variable (34).

Monitoring the training load also helps us to prevent injuries (92). Increasing training and competition time is related to greater team performance, but it also increases the number of injuries. However, higher injury rates are not associated with worse overall team

performance (17). Nevertheless, it is important to track players' exposure so as to avoid the risk of injuries because of their individual impact, even if the injuries do not affect the overall performance of the team (17). Furthermore, it has been observed that players with a lower number of decelerations and distance covered during games have a greater risk of injury. Increasing EL therefore seems to reduce injury risk (18). Athletes

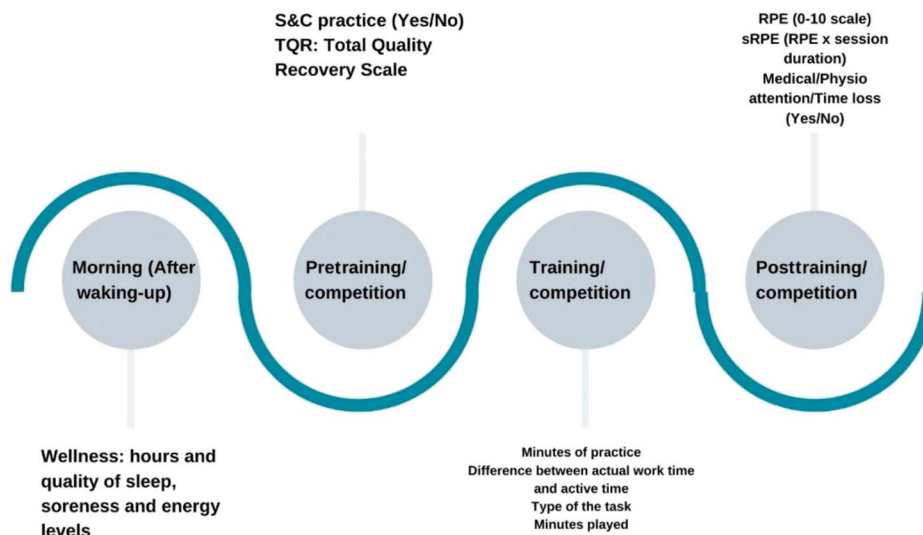


Figure 2. Suggested timeline for ecological load monitoring.

with a lower EL must be identified so that appropriate prevention strategies can be individually applied to prevent injuries (18). Planning for the season must therefore take account of its minimum and potential maximum length, having regard to the number of training sessions and the number of games (17). In elite basketball, the average number of injuries per team over the season is 23, and the data suggest that a higher number of injuries occur in training sessions (mean of 13 injuries per season) than in games (mean of 10 injuries per season). However, it has been observed that the seasonal average incidence of injuries (number of injuries per 1,000 hours) is 5 overall, 3 in training and 40 in games (17).

PRACTICAL APPLICATIONS

With a range of EL and IL measures, coaches or strength coaches should monitor the training load of their athletes. This tracking of workloads should be performed at all levels of basketball, optimizing the training process and reducing possible injury risk. To achieve this, we have to educate the other members of the coaching staff and the players about the importance of implementing these monitoring methods.

Considering the range of proposals presented here, and the applicability of ecological methods, nonavailability of technological means should not be an impediment to monitoring our athletes.

Planning should be directed from microcycle to microcycle and adjusted according to how the players respond. In this context, establishing an appropriate schedule for monitoring our players' workload day by day can be very beneficial as a way of optimizing performance and preventing injuries. In light of the results obtained, 2 scenarios are proposed for applying load control: the first is by using technology and the second is by ecological means.

If technology is available (Figure 1), it will enable us to obtain objective information about the state of our athletes and how ready they are for the training session or game. On arrival at the training facility it is proposed that Omegawave should be used to ascertain the player's internal stress level, readiness for the training session, neurological function and HR variability evaluations, and a maximal CMJ should be performed. During the session, it is proposed that WIMU PRO devices should be used to monitor EL and Polar Team devices to track IL. The EL variables that we can observe during training are high-intensity accelerations and decelerations, and in terms of IL, how long each player is in each HR zone. With this information, we can make decisions in real time. At the end of the session, we can observe the PL to find out the players' EL during training, and in terms of IL we can calculate the SHRZ and thus ascertain the load values. Another maximal CMJ could also be performed. In addition, the injuries that arise should be recorded and classified as "time loss," "physio attention," or "medical attention."

If only ecological means are available (Figure 2), an equally complete monitoring of the training process can be conducted because we have the variables we need to program and adapt the training sessions. At the start of the day, we can administer a wellness questionnaire, which provides us with information on sleep quality, stress level, fatigue, and muscular pain. This shows us each player's perception of how ready they are to face the training session or game. Before beginning the session, it is proposed that a record should firstly be taken of whether or not any physical preparation is performed. If it is, all the work performed should be recorded because this too will represent a load for the player, and the TQR test should also be administered to them.

During the session, in terms of EL, the exposure time of the session, the difference between real work time and active time, must be monitored, as it will provide us with information about how the session went, and the type of task, because depending on the purpose of the session it will be better to do one type of task rather than another. In the case of competition, the main variable to be monitored is the number of minutes played by each player. At the end of the session, we can collect the RPE, from which we can also obtain the sRPE, and record the injuries that occur, classifying them as "time loss," "physioattention," or "medical attention."

Conflicts of Interest and Source of Funding:
The authors report no conflicts of interest and no source of funding.



Aitor Piedra is a PhD Candidate in the Department of Health Sciences of the National Institute of Physical Education and Sport of Catalonia (INEFC)

and the Head of the Strength and Conditioning Department of Femení Sant Adrià (LF2) and CB Cornellà (EBA).



Javier Peña is the Director and a Senior Researcher at the Sport and Physical Activity Studies Centre (CEEAF) at the University of Vic—Central Uni-

versity of Catalonia (Spain, Barcelona) and a member of the Sports Performance Analysis Research Group (SPARG) in the same institution.



Toni Caparrós
is a professor of
the Department
of Health Sci-
ences in the
National Insti-
tute of Physical
Education and
Sport of Catalo-
nia (INEFC)

and a member of the Sports Performance
Analysis Research Group (SPARG) at
the University of Vic–Central University
of Catalonia (Barcelona, Spain).

REFERENCES

1. Ben Abdelkrim N, Castagna C, Jabri I, et al. Activity profile and physiological requirements of junior elite basketball players in relation to aerobic-anaerobic fitness. *J Strength Cond Res* 24: 2330–2342, 2010.
2. Ben Abdelkrim N, El Faza S, El Ati J. Time-motion analysis and physiological data of elite under-19-year-old basketball players during competition. *Br J Sports Med* 41: 69–75, 2007; discussion 75.
3. Anderson L, Triplett-Mcbride T, Foster C, Doberstein S, Brice G. Impact of training patterns on incidence of illness and injury during a women's collegiate basketball season. *J Strength Cond Res* 17: 734–738, 2003.
4. Aoki MS, Ronda LT, Marcelino PR, et al. Monitoring training loads in professional basketball players engaged in a periodized training program. *J Strength Cond Res* 31: 348–358, 2017.
5. Arede J, Ferreira AP, Esteves P, Gonzalo-Skok O, Leite N. Train smarter, play more: Insights about preparation and game participation in youth national team. *Res Q Exerc Sport* 91: 583–593, 2020.
6. Arruda AF, Aoki MS, Freitas CG, Coutts A, Moreira A. Planning and monitoring training loads during an in-season basketball period. *Strength Cond J* 6: 85–89, 2013.
7. Aughey RJ. Applications of GPS technologies to field sports. *Int J Sports Physiol Perform* 6: 295–310, 2011.
8. Ballesta AS, Abreuñedo J, Caparrós T. Accelerometry in basketball. Study of external load during practice. *Phys Educ Sports* 1: 100–117, 2019.
9. Banister EW, Calvert TW. Planning for future performance: Implications for long term training. *Can J Appl Sport Sci* 5: 170–176, 1980.
10. Batalla Gavalda A, Bofill AM, Montoliu Colás R, Corbi Soler F. Relationship between heart rate and the scoreboard during a relegation playoff. *J Phys Educ Sport* 132: 110–122, 2018.
11. Beneke R, Leithäuser RM, Ochentel O. Blood lactate diagnostics in exercise testing and training. *Int J Sports Physiol Perform* 6: 8–24, 2011.
12. Bengtsson H, Ekstrand J, Waldén M, Hägglund M. Match injury rates in professional soccer vary with match result, match venue, and type of competition. *Am J Sports Med* 41: 1505–1510, 2013.
13. Berkemans DM, Dalbo VJ, Fox JL, et al. The influence of different methods to determine maximum heart rate on training load outcomes in basketball players. *J Strength Cond Res* 32: 1–25, 2018.
14. Booth FW, Thomason DB. Molecular and cellular adaptation of muscle in response to exercise: Perspectives of various models. *Physiol Rev* 71: 541–585, 1991.
15. Borresen J, Ian Lambert M. The quantification of training load, the training response and the effect on performance. *Sport Med* 39: 779–795, 2009.
16. Bourdon PC, Cardinale M, Murray A, et al. Monitoring athlete training loads: Consensus statement. *Int J Sports Physiol Perform* 12: 161–170, 2017.
17. Caparrós T, Alentorn-Geli E, Myer GD, et al. The relationship of practice exposure and injury rate on game performance and season success in professional male basketball. *J Sports Sci Med* 15: 397–402, 2016.
18. Caparrós T, Casals M, Solana Á, Peña J. Low external workloads are related to higher injury risk in professional male basketball games. *J Sport Sci Med* 17: 289–297, 2018.
19. Claudino JG, Cronin J, Mezêncio B, et al. The countermovement jump to monitor neuromuscular status: A meta-analysis. *J Sci Med Sport* 20: 397–402, 2017.
20. Conte D, Kolb N, Scanlan AT, Santolamazza F. Monitoring training load and well-being during the in-season phase in national collegiate athletic association division I men's basketball. *Int J Sports Physiol Perform* 13: 1067–1074, 2018.
21. Cruz IF, Pereira LA, Kobal R, et al. Perceived training load and jumping responses following nine weeks of a competitive period in young female basketball players. *PeerJ* 6: e5225, 2018.
22. Dupont G, Nedelec M, McCall A, et al. Effect of 2 soccer matches in a week on physical performance and injury rate. *Am J Sports Med* 38: 1752–1758, 2010.
23. Fan W, Evans RM. Exercise mimetics: Impact on health and performance. *Cell Metab* 25: 242–247, 2017.
24. Foster C, Florhaug JA, Franklin J, et al. A new approach to monitoring exercise training. *J Strength Cond Res* 15: 109–115, 2001.
25. Fox JL, Stanton R, Scanlan AT. A comparison of training and competition demands in semiprofessional male basketball players. *Res Q Exerc Sport* 89: 103–111, 2018.
26. Freitas CG, Aoki MS, Arruda AF, Nakamura FY, Moreira A. Internal load, stress tolerance and upper respiratory tract infections in basketball athletes. *Rev Bras Cineantropom Perform* 15: 14, 2013.
27. Fuller CW, Junge A, Dvorak J. Risk management: FIFA's approach for protecting the health of football players. *Br J Sports Med* 46: 7–11, 2012.
28. Gabbett TJ. Incidence of injury in semi-professional rugby league players. *Br J Sports Med* 37: 36–43, 2003.
29. Gabbett TJ. The training–injury prevention paradox: Should athletes be training smarter and harder? *Br J Sport Med* 50: 273–280, 2016.
30. Gabbett TJ, Hulin BT, Blanch P, Whiteley R. High training workloads alone do not cause sports injuries: How you get there is the real issue. *Br J Sports Med* 50: 444–445, 2016.
31. Garrett J, Graham SR, Eston RG, et al. A novel method of assessment for monitoring neuromuscular fatigue in Australian rules football players. *Int J Sports Physiol Perform* 14: 598–605, 2019.
32. Gazzano F, Gabbett T. A practical guide to workload management and injury prevention in sport. *National Strength Cond Assoc* 4: 46, 2019.
33. Ghali BM, Owosye OB, Stilling C, et al. Internal and external workload in youth basketball players who are symptomatic and asymptomatic for patellar tendinopathy. *J Orthop Sport Phys Ther* 50: 402–408, 2020.
34. Gómez-Carmona CD, Bastida-Castillo A, García-Rubio J, Pino-Ortega J, Ibáñez SJ. Influence of the result on the external load demands during the official competition in basketball training. *Sports Psychol* 19: 262–274, 2019.

35. Halson SL. Monitoring training load to understand fatigue in athletes. *Sport Med* 44: 139–147, 2014.
36. Heishman AD, Curtis MA, Saliba E, et al. Noninvasive assessment of internal and external player load: Implications for optimizing athletic performance. *J Strength Cond Res* 32: 1280–1287, 2018.
37. Hopkins WG. Quantification of training in competitive sports: Methods and applications. *Sport Med* 12: 161–183, 1991.
38. Hopkins WG. A scale of magnitudes for effect statistics. 2006. pp. 67–95. Available at: <http://www.sportsci.org/resource/stats/>. Accessed October 15, 2016.
39. Hulin BT, Gabbett TJ, Blanch P, et al. Spikes in acute workload are associated with increased injury risk in elite cricket fast bowlers. *Br J Sports Med* 48: 708–712, 2014.
40. Hulin BT, Gabbett TJ, Caputi P, Lawson DW, Sampson JA. Low chronic workload and the acute:chronic workload ratio are more predictive of injury than between-match recovery time: A two-season prospective cohort study in elite rugby league players. *Br J Sports Med* 50: 1008–1012, 2016.
41. Hulin BT, Gabbett TJ, Lawson DW, Caputi P, Sampson JA. The acute:chronic workload ratio predicts injury: High chronic workload may decrease injury risk in elite rugby league players. *Br J Sports Med* 50: 231–236, 2016.
42. Hülka K, Cuberek R, Belka J. Heart rate and time-motion analyses in top junior players during basketball matches. *Acta Gymnica* 43: 27–35, 2013.
43. Huxley DJ, O'Connor D, Healey PA. An examination of the training profiles and injuries in elite youth track and field athletes. *Eur J Sport Sci* 14: 185–192, 2014.
44. Impellizzeri FM, Marcora SM, Coutts AJ. Internal and external training load: 15 years on. *Int J Sports Physiol Perform* 14: 270–273, 2019.
45. Impellizzeri FM, Rampinini E, Marcora SM. Physiological assessment of aerobic training in soccer. *J Sports Sci* 23: 583–592, 2005.
46. Jiménez-Reyes P, Pareja-Blanco F, Cuadrado-Peñañel V, et al. Jump height loss as an indicator of fatigue during sprint training. *J Sports Sci* 37: 1029–1037, 2019.
47. Jobson SA, Passfield L, Atkinson G, Barton G, Scarf P. The analysis and utilization of cycling training data. *Sport Med* 39: 833–844, 2009.
48. Kallus KW, Kellmann M, Kallus MK, Kellmann M. *The Recovery-Stress Questionnaires: User Manual RESTQ Recovery-Stress Questionnaires: User Manual*. 2016. Available at: www.pearsonassessment.de. Accessed February 11, 2020.
49. Kenttä G, Hassmén P. Overtraining and recovery. A conceptual model. *Sport Med* 26: 1–16, 1998.
50. López-Laval I, Legaz-Arrese A, George K, et al. Cardiac troponin I release after a basketball match in elite, amateur and junior players. *Clin Chem Lab Med* 54: 333–338, 2016.
51. Lupo C, Tessitore A, Gasperi L, Gomez M. Session-RPE for quantifying the load of different youth basketball training sessions. *Biol Sport* 34: 11–17, 2017.
52. Lupo C, Ungureanu AN, Frati R, et al. Player session rating of perceived exertion: A more valid tool than coaches' ratings to monitor internal training load in elite youth female basketball. *Int J Sports Physiol Perform* 5: 1–6, 2019.
53. Lyman S, Fleisig GS, Andrews JR, Osinski ED, eds. Effect of pitch type, pitch count, and pitching mechanics on risk of elbow and shoulder pain in youth baseball pitchers. *Am J Sports Med* 30: 463–468, 2002.
54. Macera CA. Lower extremity injuries in runners: Advances in prediction. *Sport Med* 13: 50–57, 1992.
55. Marcelino PR, De Arruda AF, De Oliveira R, et al. Does the level of fitness affect the magnitude of the internal training load in young basketball players? *Rev Andaluza Med Del Deporte* 6: 115–119, 2013.
56. Martin DT, Andersen MB. Heart rate-perceived exertion relationship during training and taper. *J Sports Med Phys Fitness* 40: 201–208, 2000.
57. Meeusen R, Duclos M, Foster C, et al. Prevention, diagnosis, and treatment of the overtraining syndrome: Joint consensus statement of the European College of Sport Science and the American College of Sports Medicine. *Med Sci Sports Exerc* 45: 186–205, 2013.
58. Miloski B, Aoki MS, De Freitas CG, et al. Does testosterone modulate mood states and physical performance in young basketball players? *J Strength Cond Res* 29: 2474–2481, 2015.
59. Moran LR, Hegedus EJ, Bleakley CM, Taylor JB. Jump load: Capturing the next great injury analytic. *Br J Sports Med* 53: 8–9, 2019.
60. Moreira A, McGuigan MR, Arruda AF, Freitas CG, Aoki MS. Monitoring internal load parameters during simulated and official basketball matches. *J Strength Cond Res* 26: 861–866, 2012.
61. Moreno-Pérez V, Del Coso J, Raya-González J, Nakamura FY, Castillo D. Effects of basketball match-play on ankle dorsiflexion range of motion and vertical jump performance in semi-professional players. *J Sports Med Phys Fitness* 60: 110–118, 2019.
62. Morgan WP, Brown DR, Raglin JS, O'Connor PJ, Ellickson KA. Psychological monitoring of overtraining and staleness. *Br J Sports Med* 21: 107–114, 1987.
63. Mujika I, Padilla S. Detraining: Loss of training induced physiological and performance adaptation. Part I. Short term insufficient training stimulus. *Sport Med* 30: 79–87, 2000.
64. Nederhof E, Lemmink KA, Visscher C, Meeusen R, Mulder T. Psychomotor speed: Possibly a new marker for overtraining syndrome. *Sport Med* 36: 817–828, 2006.
65. Nunes JA, Costa EC, Viveiros L, Moreira A, Aoki MS. Monitoring of internal load not basquetebol. *Rev Bras Kineanthropometry Hum Performance* 13: 67–72, 2011.
66. Nunes JA, Moreira A, Crewther BT, et al. Monitoring training load, recovery-stress state, immune-endocrine responses, and physical performance in elite female basketball players during a periodized training program. *J Strength Cond Res* 28: 2973–2980, 2014.
67. Oliveira-Da-Silva L, Sedano-Campo S, Redondo-Castán JC. Characteristics of the effort in competition in elite basketball players during the final phases of the Euroleague and the World Championship. *RICYDE Rev Int Sports Sciences* 9: 360–376, 2013.
68. Pino-Ortega J, Rojas-Valverde D, Gómez-Carmona CD, et al. Impact of contextual factors on external load during a congested-fixture tournament in elite U'18 basketball players. *Front Psychol* 10: 10–15, 2019.
69. Plews DJ, Laursen PB, Stanley J, Kilding AE, Buchheit M. Training adaptation and heart rate variability in elite endurance athletes: Opening the door to effective monitoring. *Sport Med* 43: 773–781, 2013.
70. Portes R, Navarro RM, Sosa C, Trapero JJ, Jiménez SL. Monitoring and interpreting external load in basketball: A narrative

- review. *Psicol Del Deport* 28: 119–131, 2019.
71. Pyne DB, Martin DT. Fatigue—insights from individual and team sports. *Regulat Fatigue Exercise* 28: 177–186, 2011.
72. Reina M, Mancha D, Feu S, Ibáñez SJ. Do you train like you compete? Analysis of the load in women's basketball. *J Sport Psychol* 26: 9–13, 2017.
73. Reina M, Rubio JG, Antúnez A, Ibáñez SJ. Comparison of internal and external load in official 3 vs. 3 and 5 vs. 5 female basketball competitions. *New Challeng Perspect Educ Physics Sports Recreation* 2041: 400–405, 2020.
74. Ritchie D, Hopkins WG, Buchheit M, Cordy J, Bartlett JD. Quantification of training and competition load across a season in an elite Australian football club. *Int J Sports Physiol Perform* 11: 474–479, 2016.
75. Robinson DM, Robinson SM, Hume PA, Hopkins WG. Training intensity of elite male distance runners. *Med Sci Sports Exerc* 23: 1078–1082, 1991.
76. Román MR, García-Rubio J, Feu S, Ibáñez SJ. Training and competition load monitoring and analysis of women's amateur basketball by playing position: Approach study. *Front Psychol* 9: 2689, 2019.
77. Rushall BS. A tool for measuring stress tolerance in elite athletes. *J Appl Sport Psychol* 2: 51–66, 1990.
78. Sánchez-Sánchez J, Carretero M, Valiente J, et al. Heart rate response and technical demands of different small-sided game formats in young female basketballers. *RICYDE Rev Int Sports Sciences* 14: 55–70, 2018.
79. Sánchez Ballesta A, Abruñedo J, Caparrós T. Accelerometry in basketball. Study of external load during workouts. *Apunt Educ Physics and Sports* 4 100–117, 2019.
80. Sansone P, Tessitore A, Paulauskas H, et al. Physical and physiological demands and hormonal responses in basketball small-sided games with different tactical tasks and training regimes. *J Sci Med Sport* 22: 602–606, 2019.
81. Sansone P, Tschann H, Foster C, Tessitore A. Monitoring training load and perceived recovery in female basketball. *J Strength Cond Res* 34: 2929–2936, 2018.
82. Saw AE, Main LC, Gastin PB. Monitoring the athlete training response: Subjective self-reported measures trump commonly used objective measures: A systematic review. *Br J Sports Med* 50: 281–291, 2016.
83. Scanlan AT, Fox JL, Borges NR, Dascombe BJ, Dalbo VJ. Cumulative training dose's effects on interrelationships between common training-load models during basketball activity. *Int J Sports Physiol Perform* 12: 168–174, 2017.
84. Scanlan AT, Fox JL, Poole JL, et al. A comparison of traditional and modified summated-heart-rate-zones models to measure internal training load in basketball players. *Meas Phys Educ Exerc Sci* 22: 303–309, 2018.
85. Scanlan AT, Tucker PS, Dascombe BJ, et al. Fluctuations in activity demands across game quarters in professional and semiprofessional male basketball. *J Strength Cond Res* 29: 3006–3015, 2015.
86. Scanlan AT, Wen N, Tucker PS, Dalbo VJ. The relationships between internal and external training load models during basketball training. *J Strength Cond Res* 28: 2397–2405, 2014.
87. Schelling X, Torres-Ronda L. Conditioning for basketball: Quality and quantity of training. *Strength Cond J* 35: 89–94, 2013.
88. Schelling X, Torres L. Accelerometer load profiles for basketball-specific drills in elite players. *J Sports Sci Med* 15: 585, 2016.
89. Snyder A, Jeukendrup A, Hesselink M, Kuipers H, Foster C. A physiological/psychological indicator of over-reaching during intensive training. *Int J Sports Med* 14: 29–32, 1993.
90. Soligard T, Schwelnus M, Alonso JM, et al. **How much is too much? (part 1) international olympic committee consensus statement on load in sport and risk of injury.** *Br J Sports Med* 50: 1030–1041, 2016.
91. Stojanović E, Stojiljković N, Stanković R, et al. Recreational basketball small-sided games elicit high-intensity exercise with low perceptual demand. *J Strength Cond Res* 1: 10, 2019.
92. Svilar L, Castellano J, Jukic I. Load monitoring system in top-level basketball team: Relationship between external and internal training load. *Kinesiology* 50: 25–33, 2018.
93. Svilar L, Castellano J, Jukic I, Casamichana D. Positional differences in elite basketball: Selecting appropriate training-load measures. *Int J Sports Physiol Perform* 13: 947–952, 2018.
94. Twist C, Highton J. Monitoring fatigue and recovery in rugby league players. *Int J Sports Physiol Perform* 8: 467–474, 2013.
95. Vallés-Ortega C, Fernández-Ozcorta EJ, Fierro-Suero S. Fatigue-recovery pattern in a high-density competitive competition in junior women's basketball. *Quad Psicol Del Deport* 17: 183–188, 2017.
96. Vaquera A, Suárez-Iglesias D, Guiu X, et al. Physiological responses to and athlete and coach perceptions of exertion during small-sided basketball games. *J Strength Cond Res* 32: 2949–2953, 2018.
97. Vazquez-Guerrero J, Reche X, Cos F, Casamichana D, Sampaio J. Changes in external load when modifying rules of 5-on-5 scrimmage situations in elite basketball. *J Strength Cond Res* 1: 10–15, 2018.
98. Vázquez-Guerrero J, Suarez-Arrones L, Gómez DC, Rodas G. Comparing external total load, acceleration and deceleration outputs in elite basketball players across positions during match play. *Kinesiology* 50: 228–234, 2018.
99. Vivek M. *Elite HRV: Top Heart Rate Variability App, Monitors, and Training.* Available at: <https://elitehrv.com/>. Accessed March 31, 2020.
100. Xevi S. Monitoring of athletes and players for team and individual sports. Available at: <https://www.quanter.io/es.html>. Accessed March 31, 2020.
101. xStart. Homepage—myCoach RPE. Available at: <http://www.mycoachrpe.com/?lang=es>. Accessed March 31, 2020.
102. xStart. Readiness: Wellness monitoring by carlos balsalobre. Available at: <https://appadvice.com/app/readiness-wellness-monitoring/1437719482>. Accessed March 31, 2020.